

Experiment 'p_2_7_circle_properties' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_7_circle_properties

Natural language statement: null

Method signature: p_2_7_circle_properties(radius: real) returns (circumference: real, area: real, volume: real, surface_area: real)

Ensures

- `circumference == 2.0 * 3.14159 * radius`
- `area == 3.14159 * radius * radius`
- `volume == (4.0 / 3.0) * 3.14159 * radius * radius * radius`
- `surface_area == 4.0 * 3.14159 * radius * radius`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_7_circle_properties(radius: real)` returns `(circumference: real, area: real, volume: real, surface_area: real)`

The method should respect the following contract:

`ensures circumference == 2.0 * 3.14159 * radius, ensures area == 3.14159 * radius * radius, ensures volume == (4.0 / 3.0) * 3.14159 * radius * radius * radius, ensures surface_area == 4.0 * 3.14159 * radius * radius`

Produce and show only the Dafny body of this method, including the curly braces that surround it. Do not show the signature nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2mr64ILOU3NnNSeSyRSxylFQ5m8

Model: gpt-5.5-2026-04-23

Created at: 1780581157

Finish reason: stop

Usage: 408 tokens in, and 200 tokens out

```
{
  circumference := 2.0 * 3.14159 * radius;
  area := 3.14159 * radius * radius;
  volume := (4.0 / 3.0) * 3.14159 * radius * radius * radius;
  surface_area := 4.0 * 3.14159 * radius * radius;
}
```

Final Program

```
method p_2_7_circle_properties(radius: real) returns (circumference: real, area: real,
  volume: real, surface_area: real)
  ensures circumference == 2.0 * 3.14159 * radius
  ensures area == 3.14159 * radius * radius
  ensures volume == (4.0 / 3.0) * 3.14159 * radius * radius * radius
  ensures surface_area == 4.0 * 3.14159 * radius * radius
{
```

```
circumference := 2.0 * 3.14159 * radius;  
area := 3.14159 * radius * radius;  
volume := (4.0 / 3.0) * 3.14159 * radius * radius * radius;  
surface_area := 4.0 * 3.14159 * radius * radius;  
}
```

Total Token Usage

Input tokens: 408

Output tokens: 200

Reasoning tokens: 102

Sum of ‘total tokens’: 608

Experiment Timings

Overall Experiment started at 1780581156299, ended at 1780581191131, lasting 34832ms (34.83 seconds)

Iteration #1 started at 1780581156305, ended at 1780581191131, lasting 34826ms (34.83 seconds)